

Question Number	Answer	Mark
6(a)(i)	The only correct answer is C Y <i>A is not correct because the xylem is Y</i> <i>B is not correct because the xylem is Y</i> <i>D is not correct because the xylem is Y</i>	(1)

Question Number	Answer	Mark
6(a)(ii)	The only correct answer is B W <i>A is not correct because the phloem is W</i> <i>C is not correct because the phloem is W</i> <i>D is not correct because the phloem is W</i>	(1)

Question Number	Answer	Additional guidance	Mark
6(a)(iii)	An answer that makes reference to the following points: • millipede gains {water / mineral ions} from xylem (1) • millipede gains {sucrose / organic solutes / organic substances / amino acids} from phloem (1) • appropriate description of use of gained material (1)	IGNORE xylem contains {water / mineral ions} IGNORE glucose IGNORE phloem contains {sucrose / organic solutes} IGNORE nutrients / sugar / water from phloem e.g. {sucrose / glucose} used for respiration / amino acids to allow growth / energy source for growth / water for hydrolysis reactions / water for solvent / water to prevent dehydration / calcium ions for exoskeleton ACCEPT more efficient than getting from other sources IGNORE nitrates IGNORE increase survival of millipede IGNORE produce energy	(3)

Question Number	Answer	Additional guidance	Mark
6(b)(i)	An answer that makes reference to the following points: • (hard exoskeleton) for {defence / protection} from {predators / damage} / protection of {tissues / organs} (1) • (poison glands) kill {predators / prey} / for {defence / protection / deterrent} from predators (1)	ACCEPT for muscle attachment to allow movement ACCEPT to protect soft insides / withstand high pressure ACCEPT it can't be eaten ACCEPT to prevent {dehydration / loss of water} IGNORE protection unqualified IGNORE camouflage / survival / help make holes IGNORE for movement unqualified IGNORE to digest plants	(2)

Question Number	Answer	Additional guidance	Mark
6(b)(ii)	An answer that makes reference to the following points: - analysis of {genotype / phenotype} of both species of millipede (1) - identification of the number of {similarities / differences} between the two species (1)	e.g. looking at {DNA/mRNA/amino acid/ gene / allele} {sequences / structure} / protein structure / named phenotypic characteristics / proteomics / molecular phylogeny IGNORE genetic mapping ACCEPT the more similarities the more closely related / converse IGNORE compare sequences unqualified	(2)

Question Number	Answer	Additional guidance	Mark
6(c)	An answer that makes reference to three of the following points: <u>Similarities: max 2</u> • correct similarity (1) • second correct similarity (1) <u>Differences: max 2</u> • correct difference (1) • second correct difference (1)	e.g. both contain a cell membrane both contain cell wall both contain ribosomes both contain cytoplasm both contain glycogen granules both contain {DNA / genetic material} REJECT incorrect similarities e.g. differences in cell wall / nucleus / nucleoid / mitochondria / type of ribosome / pilus / flagellum / DNA format REJECT incorrect differences	(3)